Properties of the Fourier Transform

Copyright © Barry Van Veen 2014

AllSignalProcessing.com

Key Concepts

1) The properties of the Fourier transform help us develop insight into the nature of frequency-domain representations for signals.

2) The Fourier transform is a linear operation so superposition holds.

$$ax(t) + by(t) \overset{FT}{\longleftrightarrow} aX(\Omega) + bY(\Omega)$$

The Fourier transform of a weighted sum of signals is the weighted sum of the individual Fourier transforms.

3) Symmetry for real valued $x(t)$:

a) Conjugate symmetry: $X(-\Omega) = X^*(\Omega)$.

b) $\text{Real}\{X(\Omega)\} = \text{Real}\{X(-\Omega)\}$. The real part has even symmetry.

c) $\text{Imag}\{X(\Omega)\} = -\text{ Imag}\{X(-\Omega)\}$. The imaginary part has odd symmetry.

d) $|X(\Omega)| = |X(-\Omega)|$. The magnitude has even symmetry.

e) $\arg\{X(\Omega)\} = -\arg\{X(-\Omega)\}$. The phase has odd symmetry.

4) Time shift:

$$x(t - t_o) \overset{FT}{\longleftrightarrow} e^{j\Omega t_o} X(\Omega)$$

A shift in time introduces a phase shift that is a linear function of frequency.

5) Frequency shift:

$$e^{j\gamma t} x(t) \overset{FT}{\longleftrightarrow} X(\Omega - \gamma)$$

Multiplication by complex sinusoid in time introduces a frequency shift.

6) Scaling

$$x(at) \overset{FT}{\longleftrightarrow} \frac{1}{|a|} X\left(\frac{\Omega}{a}\right)$$

Compressing ($|a| < 1$) the signal in time corresponds to expanding it in frequency and vice versa.

7) Differentiation in time:

$$\frac{d}{dt}x(t) \overset{FT}{\longleftrightarrow} j\Omega X(\Omega)$$

Differentiation in time corresponds to multiplication by $j\Omega$.

8) Parseval's Theorem:

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(\Omega)|^2 d\Omega$$

The energy in the time- and frequency-domain representations is identical.

9) Convolution-multiplication

$$x(t) * y(t) = \int_{-\infty}^{\infty} x(\tau) y(t-\tau) d\tau \overset{FT}{\longleftrightarrow} X(\Omega) Y(\Omega)$$

Convolution of time-domain signals corresponds to multiplication of the corresponding Fourier transforms.

10) Multiplication-convolution

$$x(t)y(t) \overset{FT}{\longleftrightarrow} \frac{1}{2\pi}X(\Omega) * Y(\Omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\Gamma)Y(\Omega - \Gamma)d\Gamma$$

Multiplication of time-domain signals corresponds to convolution of the corresponding Fourier transforms.

Video Screenshots

Fourier Transform Properties: give insight

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\Omega) e^{j\Omega t} d\Omega \quad \overset{FT}{\longleftrightarrow} \quad X(\Omega) = \int_{-\infty}^{\infty} x(t) e^{-j\Omega t} dt$$

1) Linearity

$$a x(t) + b y(t) \quad \overset{FT}{\longleftrightarrow} \quad a X(\Omega) + b Y(\Omega)$$

2) Symmetry. Suppose $x(t)$ is real

$$X^*(\Omega) = \left[\int_{-\infty}^{\infty} x(t) e^{-j\Omega t}\, dt\right]^* = \int_{-\infty}^{\infty} x^*(t)\, e^{j\Omega t} dt$$

$$= \int_{-\infty}^{\infty} x(t)\, e^{-j(-\Omega)t} dt = X(-\Omega)$$

$\Rightarrow$ $\mathrm{Real}\{X(\Omega)\}$ is even $\quad$ $|X(\Omega)|$ is even

$\mathrm{Imag}\{X(\Omega)\}$ is odd $\quad$ $\arg\{X(\Omega)\}$ is odd

3) Time shift. $y(t) = x(t - t_0)$

$$Y(\Omega) = \int_{-\infty}^{\infty} x(t-t_0)\, e^{-j\Omega t}\, dt = \int_{-\infty}^{\infty} x(v)\, e^{-j\Omega(v+t_0)}\, dv$$

$$= e^{-j\Omega t_0}\, X(\Omega)$$

$$x(t-t_0) \overset{FT}{\longleftrightarrow} e^{-j\Omega t_0} X(\Omega)$$

$$|Y(\Omega)| = |X(\Omega)|$$

$$\arg\{Y(\Omega)\} = \arg\{X(\Omega)\} - \Omega t_0$$

4) Frequency Shift

$$e^{j\gamma t}\, x(t) \overset{FT}{\longleftrightarrow} X(\Omega - \gamma)$$

Ex. $\cos(100t)\, e^{-t}\, u(t) \overset{FT}{\longleftrightarrow} \frac{0.5}{1 + j(\Omega - 100)} + \frac{0.5}{1 + j(\Omega + 100)}$

5) Scaling

$$x(at) \overset{FT}{\longleftrightarrow} \frac{1}{|a|} X\left(\frac{\Omega}{a}\right)$$

$a > 1 \Rightarrow$ compress (speed up) signal in time $\Rightarrow$ expand in frequency

- inverse relation between time and frequency extent

4) Differentiation. $y(t) = \frac{d}{dt} x(t)$

$$y(t) = \frac{d}{dt}\left\{\frac{1}{2\pi}\int_{-\infty}^{\infty} X(\Omega)\, e^{j\Omega t}\, d\Omega\right\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} (X(\Omega)\, j\Omega)\, e^{j\Omega t}\, d\Omega$$

$x(t)$ $\qquad$ $Y(\Omega)$

$\frac{d}{dt}x(t) \overset{FT}{\longleftrightarrow} j\Omega X(\Omega)$

amplifies high frequencies

5) Parseval's Theorem

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(\Omega)|^2 d\Omega$$

Find $\beta = \int_{-\infty}^{\infty} \frac{4}{|j\Omega + 2|^2} d\Omega$. $\qquad 2e^{-2t}u(t) \overset{FT}{\longleftrightarrow} \frac{2}{j\Omega + 2}$

$$\beta = 2\pi \int_{-\infty}^{\infty} 4e^{-4t}u(t)dt = 2\pi\int_{0}^{\infty} 4e^{-4t}dt = 2\pi\left[-e^{-4t}\right]_0^{\infty} = 2\pi$$

Convolution – Multiplication

$$\int_{-\infty}^{\infty} x(\tau)\,y(t-\tau)\,d\tau = x(t)*y(t) \quad \xleftrightarrow{FT} \quad X(\Omega)\,Y(\Omega)$$

Find $y(t) = x(t) * h(t)$

$$x(t) = \frac{\sin(4t)}{\pi t}, \quad h(t) = \frac{\sin(2t)}{\pi t}$$

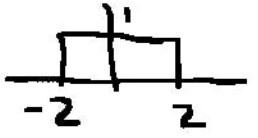
$$Y(\Omega) = X(\Omega)\,H(\Omega) \qquad X(\Omega) = \begin{cases} 1 & |\Omega| < 4 \\ 0 & |\Omega| > 4 \end{cases}$$

$$H(\Omega) = \begin{cases} 1 & |\Omega| < 2 \\ 0 & |\Omega| > 2 \end{cases} \Rightarrow Y(\Omega) = H(\Omega)$$

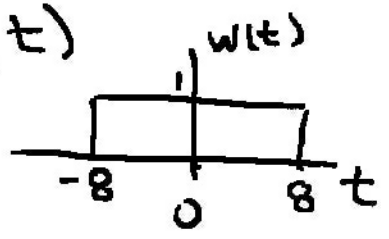
$$y(t) = h(t) = \sin(2t)/\pi t$$

Multiplication – Convolution

$$x(t)\,y(t) \quad \xleftrightarrow{FT} \quad \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\nu)\,Y(\nu-\Omega)\,d\nu = \frac{1}{2\pi}\,X(\Omega)*Y(\Omega)$$

Truncation $h(t) = \dfrac{\sin(2t)}{\pi t} \xleftrightarrow{FT}$ 

; $g(t) = h(t)\,w(t)$



$$G(\Omega) = \frac{1}{2\pi}H(\Omega)*W(\Omega) = \frac{1}{2\pi}\int_{-2}^{2} \frac{2\sin(8(\nu-\Omega))}{\nu-\Omega}\,d\nu$$

For more great educational signal-processing content:

AllSignalProcessing.com

- Intuitive, easy-to-follow style
- Organization that saves you time
- Answers to your questions
- Learn at your pace
- Award winning instruction